Assessment Brief

Module information

Module Name	Computational Mathematics and Data Analytics
Module Code	M-CSC005U1
Module Leader	Kenneth Nweke

Assessment Information 1

Assessment Title	Mathematical and Data Analytics Group Project
Assessment Type	Group Project (Course Work)
Due Date & Time	Monday 29/06/2026; 14:00
Feedback Date	Monday; 13/07/2026; 09:00
Intended Learning Outcomes *(delete rows below as required)*	2. Construct and evaluate mathematical and statistical models to represent and interpret real-world computing scenarios. 4. Communicate analytical results effectively and ethically through structured technical reports, visual representations, and collaborative projects.
Weighting towards final module mark (%)	60%
Word Count/Time allocation	Report + Individual Reflective Statement. 3000 + 500 Words

AI Category	Description	Examples *Please provide further information in the task outline*	In Use *Select one*
A	Generative AI tools may not be used due to the format and purpose of the assessment.	Such assessments may include: • Exams • Online tests • Vivas • Practical assessments	☐
B	Generative AI tools may be used in a limited assistive role for this assessment, but AI generated outputs should not be submitted within the assessment.	In this category, AI may be used to provide initial suggestions on: • Essay structure • Recommendations for further reading • Creative inspirations	☒
C	Generative AI tools may be used to create specific outputs to inform aspects of the assessment, with appropriate referencing and citation.	With the overt use of AI, the focus of assessment should involve critique and/or reflection on the generated outputs. AI generated content submitted with the assessment must be clearly identified as such.	☐

Task Outline

Task Overview

Part A: Group (3-4 students) Project Report and Code Submission (50% of assessment)

Your group will select one computational or data-driven problem from the approved project list (or propose your own with tutor approval by ***Week 8 23/03/2026 – 27/03/2026***) and develop a mathematical model or data analytics solution using appropriate computational techniques.

Project Deliverables:

- **Technical Report** (2,500-3,000 words)
    1. Executive Summary
    2. Problem Definition and Context
    3. Mathematical/Statistical Methodology
    4. Implementation Approach
    5. Results and Analysis
    6. Ethical Considerations
    7. Conclusions and Recommendations
    8. References (Harvard format)
- **Code Repository**
    1. Well-commented Python/MATLAB/SQL Code
    2. README file with setup instructions
    3. Sample datasets (if applicable)
- **Visual Outputs**
    1. Minimum 4 high-quality visualizations
    2. Charts, graphs, or network diagrams as appropriate
    3. Clear labelling and professional presentation

Indicative Project Topics:

- **Network Analysis:** Model and analyse social network connectivity using graph theory
- **Algorithm Optimization:** Compare sorting/searching algorithms with complexity analysis
- **Database Design:** Design and implement a relational database for a real-world scenario with SQL queries
- **Predictive Modelling:** Use statistical regression to predict trends in computing datasets
- **Data Analytics Case Study:** Clean, analyse, and visualize a messy real-world dataset
- **Simulation Study:** Model a probabilistic computing scenario (e.g., server queue management)
- **Cryptography Application:** Implement and analyse modular arithmetic in hash functions

Part B: Individual Reflective Statement (10% of assessment)

Each student must submit a 500-word reflective statement addressing:

- Your specific contributions to the project (tasks, coding, research, writing)
- Challenges encountered and how you overcame them
- Mathematical or analytical concepts you found most valuable
- How group collaboration enhanced your learning
- Areas for personal development

Assessment Submission Guidelines

- **Code:** Zipped folder or GitHub repository link
- **Report + *Individual Reflective Statement*,** must be submitted as a single PDF clearly labelled with your student ID number in the VLE, Turnitin Link.
- The assignment should be formatted in 12pt font (Arial or Times New Roman), double-spaced.
- The assignment sections should follow academic formatting.

Academic Integrity and Group Work Guidelines

- All group members must contribute meaningfully to the project
- Code must be original or properly attributed (libraries excepted)
- AI tools (e.g., ChatGPT, Claude, Copilot) may be used for code debugging and learning but not for generating complete solutions - document any AI assistance in your README
- Suspected collusion or plagiarism will be investigated under university regulations
- Groups experiencing significant dysfunction should contact the module leader bimidiately Week 17

You must declare the word count for the assessment (excluding reference lists, bibliographies, tables and the title of tables, graphs, and appendices) at the end of the written submission, situated before the reference list and bibliography. Falsification of the wordcount will be considered as academic malpractice.

Marks and Feedback

Marking Rubric: Group Project Report and Code (50%)

Grade	**Mark**	**Characteristic**	**Mathematical Rigor** (25%)	**Implementation Quality** (25%)	**Analysis & Communication** (25%)	**Ethical Awareness** (15%)	**Professional Presentation** (10%)
A	**70-100**	**Excellent Pass**	Sophisticated application of mathematical concepts with clear understanding of underlying theory	Professional, well-documented code that is efficient and fully functional	Insightful analysis with excellent visual communication and clear technical writing	Thoughtful consideration of ethical implications with specific examples	Exceptionally well-structured with flawless formatting and references
B	**60-69**	**Good Pass**	Sound application of mathematical methods with mostly accurate reasoning	Good quality code with adequate documentation and mostly functional	Good analysis with effective visualizations and clear writing	Good awareness of ethical issues with some discussion	Well-organized with good formatting and proper
C	**50-59**	**Clear Pass**	Basic mathematical approach with some understanding shown	Functional code with basic documentation and some issues	Adequate analysis with acceptable visuals and understandable writing	Basic acknowledgment of ethical considerations	Adequately structured with acceptable formatting
D	**40-49**	**Threshold/Satisfactory Pass**	Limited mathematical understanding with significant gaps	Code runs with significant issues or poor documentation	Limited analysis with weak visualizations or unclear writing	Minimal ethical discussion	Poor structure or incomplete formatting
E	**0-39**	**Needs Improvement**	Limited mathematical understanding with significant gaps	Non-functional code or missing implementation	Poor or missing analysis and communication	No ethical consideration	Unprofessional or missing key elements

Marking Rubric: Individual Reflection (10%)

Grade	Mark	Characteristic	Self-Awareness (50%)	Evidence of Contribution (50%)
A	70-100	**Excellent Pass**	Deep, critical reflection on learning and contribution	Specific, detailed evidence of substantial contribution
B	60-69	**Good Pass**	Good reflection with clear self-assessment	Clear evidence of good contribution
C	50-59	**Clear Pass**	Basic reflection with some self-awareness	Adequate evidence of contribution
D	40-49	**Threshold/Satisfactory Pass**	Superficial reflection with limited insight	Vague or limited evidence
E	0-39	**Needs Improvement**	Minimal or no meaningful reflection	No evidence of contribution

Written Assessments:

The upper word limit specified for this assessment may be exceeded by up to 10% without penalty. There is no lower limit but you should consider how you are taking advantage of the wordcount to demonstrate all of the learning outcomes. Where the word count exceeds 10%, the assessment will be marked and the following penalties will be applied.

- *0% - 10% No penalty*
- *11%- 30% 10 marks will be deducted, or the pass mark awarded (whichever is the higher)*
- *31% + Maximum mark awarded is a pass mark*

Assessment Information 2

Assessment Title	Mathematics and Data Analytics In-class Test
Assessment Type	In-class Test (PRA EX)
Due Date & Time	Monday 27/07/2026; 17:00
Feedback Date	Monday; 10/08/2026; 09:00
Intended Learning Outcomes *(delete rows below as required)*	1. Apply discrete mathematical and algebraic concepts to solve computational problems. 3. Design and query simple relational databases using SQL and apply data manipulation tools to prepare, clean, and visualise datasets.
Weighting towards final module mark (%)	40%
Duration	2 Hours

Assessment Brief

This supervised, closed-book assessment tests your ability to independently apply mathematical, statistical, and computational reasoning to solve problems across the module's core topics.

Assessment Structure:

Section A: Discrete Mathematics and Logic (30 marks)

- Propositional and predicate logic
- Set theory and Boolean algebra
- Counting principles and combinatorics
- Graph theory fundamentals

Section B: Computational Mathematics (30 marks)

- Number systems and data representation
- Algorithmic complexity and Big O notation
- Recursion and recurrence relations
- Numerical methods (root finding, error analysis)

Section C: Data Analytics and SQL (25 marks)

- Relational database design
- SQL queries (SELECT, JOIN, GROUP BY, aggregate functions)
- Data interpretation and statistical reasoning
- Probability concepts in computing

Section D: Applied Problem Solving (15 marks)

- One extended problem requiring integration of multiple concepts
- May include computational thinking, modelling, or data analysis scenario

Assessment Format

- **Total marks:** 100 (converted to 40% of module grade)
- **Question types:** MCQ, Short answer, calculation problems, SQL queries, extended response
- **Materials provided:** Formula sheet (distributed in Week 23), graph paper if needed

Sample Question Types:

1. Construct a truth table for the logical expression: $(P \vee Q) \rightarrow (\neg R \wedge P)$
2. Analyse the time complexity of the following algorithm and justify your answer using Big O notation…
3. Write a SQL query to find the top 5 customers by total purchase value from tables: Customers, Orders, Order Details
4. A dataset has 1000 records. Calculate the probability that…

Assessment Criteria

Grade Band	Descriptor
70–100%	Consistently accurate solutions with clear working. Demonstrates deep understanding of mathematical principles. Excellent problem-solving skills with insightful approaches. Minimal errors.
60–69%	Mostly accurate solutions with good understanding shown. Sound problem-solving approach. Minor errors that do not undermine overall understanding.
50–59%	Adequate solutions with reasonable understanding. Some errors in application or calculation. Problem-solving approach is generally appropriate.
40–49%	Basic understanding demonstrated but with significant gaps. Multiple errors or incomplete solutions. Limited problem-solving ability shown.
0–39%	Minimal understanding with fundamental errors. Incomplete or incorrect solutions throughout. Unable to apply concepts appropriately.

Preparation and Revision

- **Mock TCA:** 07/July/2026 (formative feedback provided)
- **Formula Sheet:** Distributed
- **Q&A Clinic:** Every week

Recommended Study Approach:

1. Review weekly formative problem sheets and solutions
2. Consolidate understanding of key formulae and methods
3. Attend all revision sessions and clinics
4. Form study groups to work through practice problems